27/01/24 A/S

USN [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] **[23CSBSMA12]**

FIRST SEMESTER B.Tech EXAMINATION, JANUARY - 2024
INTRODUCTORY TOPICS IN STATISTICS, PROBABILITY AND CALCULUS

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.
ii. Use Statistical tables if necessary

ANSWER ALL QUESTIONS

		PART-A	**05 X 02=10**		
1.	**a.**	Find $\frac{dy}{dx}$ if $x = at^2$ & $y = 2at$	L-2	CO1	**02**
	b.	Evaluate $\int_0^1 e^{2x}dx$	L-2	CO2	**02**
	c.	Given median is 20.6 and mode is 26. Find mean.	L-2	CO3	**02**
	d.	What is the probability of getting an ace or a spade, when a card is from a deck?	L-2	CO4	**02**
	e.	If X is Uniform between -2 and 6 then find mean of X.	L-2	CO5	**02**
		PART-B	**06 X 05=30**		
2.	**a.**	Find k for which $f(x) = \begin{cases} \frac{x^4-256}{x-4} & when\ x \neq 4 \\ k & when\ x = 4 \end{cases}$ is continuous at $x = 4$.	L-2	CO1	**05**
		OR			
	b.	If $x^m y^n = (x+y)^{m+n}$. Show that $\frac{dy}{dx} = \frac{y}{x}$	L-2	CO1	**05**
3.	**a.**	Evaluate $\int \frac{3}{(x+1)(x+2)}dx$	L-3	CO2	**05**
		OR			
	b.	Find the area bounded by the parabola $y^2 = 4x$ and the line $x - y = 0$.	L-3	CO2	**05**
4.	**a.**	Represent the following data by a histogram:	L-3	CO3	**05**

Variable	100-110	110-120	120-130	130-140	140-150	150-160	160-170
Frequency	11	28	36	49	33	20	8

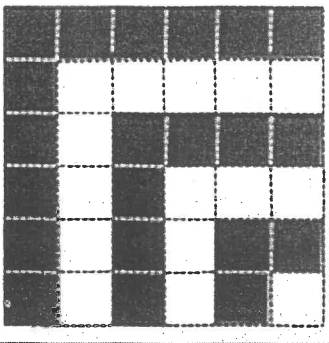
		OR			
	b.	From the following data compute the value of harmonic mean:	L-3	CO3	**05**

Class interval	10-20	20-30	30-40	40-50	50-60
Frequency	4	6	10	7	3

5.	**a.**	One bag contains 4 white balls and 3 black balls, and a second bag contains 3 white balls and 5 black balls. One ball is drawn from the first bag and placed unseen in the second bag. What is the probability that a ball now drawn from the second bag is black?	L-3	CO4	**05**
		OR			
	b.	A bag contains 40 tickets numbered 1,2,3,. . . ., 40, of which four are drawn at random and arranged in ascending order $(t_1 < t_2 < t_3 < t_4)$. Find the probability of t_3 being 25.	L-3	CO4	**05**
6.	**a.**	The Probability that a man aged 60 will live to be 70 is 0.65. What is the probability that out of 10 men, now aged 60, (i) exactly 9 will to be 70, (ii) at most 9 will live to be 70?	L-3	CO5	**05**
		OR			
	b.	In a Poisson distribution $3P(X = 2) = P(X = 4)$. Find the parameter 'm'.	L-3	CO5	**05**
7.	**a.**	Following table gives the birth rate per thousand of different countries over a certain period. Represent the following data by using a suitable diagram.	L-3	CO3	**05**

Country	Birth Rate
India	33
Germany	16
U.K.	20
China	40
Sweden	15

		OR			
	b.	A square game board has blue and white stripes of equal width as shown. What is the chance that a dart thrown at the board will land on a white stripe?	L-3	CO3	**05**



		PART-C	**06 X 10=60**		

8.	**a.**	i. If $xy + 6y = 2x$ then show that $\frac{d^2y}{dx^2} = -\frac{24}{(x+6)^3}$ ii. If $x = a\theta$ and $y = \frac{a}{\theta}$, then prove that $\frac{dy}{dx} + \frac{y}{x} = 0$	L-3	CO1	**10**
		OR			
	b.	Find the maxima and minima of the function $f(x) = 3x^3 - 9x^2 - 27x + 30$.	L-3	CO1	**10**
9.	**a.**	Evaluate $\int \frac{1}{x[(\log x)^2 - 3\log x + 2]} dx$	L-3	CO2	**10**
		OR			
	b.	Evaluate $\iint xy\, dx\, dy$ over the positive quadrant of the circle $x^2 + y^2 = a^2$	L-3	CO2	**10**
10.	**a.**	From the following data compute arithmetic mean by direct and short-cut method:	L-3	CO3	**10**

Marks	0-10	10-20	20-30	30-40	40-50	50-60
No. of Students	5	10	25	30	20	10

		OR			
	b.	From the following data of weight of 122 persons determine the modal weight.	L-3	CO3	**10**

Weight (in lbs)	100-110	110-120	120-130	130-140	140-150	150-160	160-170	170-180
No. of persons	4	6	20	32	33	17	8	2

11.	**a.**	A manufacturing firm employs three analytical plants for the design and development of a particular product. For cost reasons, all three are used at varying times. In fact, plants 1, 2, and 3 are used for 30%, 20%, and 50% of the products, respectively. The defect rate is different for the three procedures as follows: $P(D\|P_1) = 0.01, P(D\|P_2) = 0.03$ and $P(D\|P_3) = 0.02$, where $P(D\|P_j)$ is the probability of a defective product, given plan j. If a random product was observed and found to be defective, which plant was most likely used and thus responsible?	L-3	CO4	**10**
		OR			
	b.		L-3	CO4	**10**

Let X be a discrete random variable with the following probability distribution x	0	1	2	3
$f(x)$	0.15	0.38	0.10	0.01

Find

i. E(X)

ii. Variance

iii. Standard Deviation

12. a. The average monthly sales of 5000 firms are normally distributed. Its mean and standard deviation are Rs. 36000 and Rs. 10000 respectively. Find

i) The number of firms the sales of which are over Rs. 40000

ii) The percentage of firms the sales of which will be between Rs. 38500 and Rs. 41000.

iii) The number of firms the sales of which will be between Rs. 30000 and Rs. 40000.

L-3 CO5 **10**

OR

b. A die is thrown 60 times and a frequency distribution for number appearing on face (x) is given by:

X	**1**	**2**	**3**	**4**	**5**	**6**
frequency	15	6	4	7	11	17

L-3 CO5 **10**

13. a. Find the volume of the tetrahedron $\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$ and the coordinate planes.

L-3 CO2 **10**

OR

b. Suppose that in a public opinion survey answer to questions-

i. Do you drink?

ii. Are you in favor of local option on sale of liquor?

Were as given in table. Can you infer that opinion on local option is dependent on whether or not an individual drinks?

Question b)	**Question a)**		**Total**
	Yes	No	
Yes	56	31	87
no	18	6	24
Total	74	37	n= 111

L-3 CO5 **10**